# OLOS: Open Loyalty Operating System

*A vendor-neutral, privacy-preserving interoperability protocol for customer loyalty programs.*

**Status: Public Proposal / Open Protocol Specification.** Published by the OLOS Architecture Working Group for technical review by payments, security, and distributed-systems practitioners. Not production software or a certified standard.

## Why This Matters

Modern commerce runs on shared, neutral protocols that no single company owns: SMTP for email, HTTP for the web, and card/interbank networks for payments. Each succeeded by giving independent participants a common rail to exchange value or information — without forcing them to merge into one platform.

Customer loyalty never got its version of this. Every program — airline miles, retailer points, hotel status — keeps its own isolated ledger, with no common settlement layer connecting one program's value to another's. The result is a structural, not marketing, problem: merchants carry unredeemed-liability balances they can't easily resolve, consumers accumulate small scattered balances that are rarely worth redeeming, and there's no shared rail for cross-program exchange the way there is for cross-bank payments.

*(Note: specific liability and redemption-rate figures are omitted here pending a named, citable industry source — add back with a citation if available, rather than stating as settled fact.)*

## Why Prior Attempts Fell Short

- **Coalition programs** (pooled-currency schemes) require merchants to surrender their direct customer relationship and branding to a central operator — a real structural cost, not just an adoption hurdle.
- **Point-exchange intermediaries** extract meaningful spreads and typically settle in slow, batch-reconciled cycles rather than anything close to real time.
- **Blockchain/token-based loyalty proposals** generally failed on UX and volatility grounds — and, separately, most loyalty liabilities don't actually need a distributed ledger's properties (public verifiability by untrusted parties) to solve the interoperability problem.

OLOS takes a different approach: sovereign programs keep their own currency, rules, and customer relationships, and exchange value through a shared, neutral protocol layer — not a shared ledger or shared token.

## How It Actually Works

OLOS is a **message-envelope protocol and settlement specification**, not a blockchain or token system. It defines five logical planes:

| Plane | Responsibility |
|---|---|
| **Ingestion / Data** | Validates and routes signed transaction envelopes at the edge (POS gateways) |

| | |
|---|---|
| **Governance / Control** | Trust Registry — source of truth for node keys, escrow status, hardware attestation |
| **Compute** | Deterministic rules engine evaluating issuance and redemption logic |
| **Privacy / Cryptographic** | Derives Merchant-Specific Pseudonyms so no raw customer PII crosses the network |
| **Ledger / Clearing** | Nets multi-party obligations into signed settlement manifests — non-fiat asset netting, not on-chain settlement |

Core technical commitments: **deterministic integer-only math** (no floating point, to eliminate cross-runtime rounding disputes), **Ed25519-signed message envelopes**, **HSM/TEE-backed identity blinding**, and **bounded offline authorization** for disconnected point-of-sale terminals — all detailed in the full technical specification.

## What's in This Repository

| File | Covers |
|---|---|
| **OLOS-Technical-Specification-v2_0_2.pdf** | The core normative protocol spec — start here |
| **OLOS-V3-Protocol.pdf** | Offline double-spend security model and threat analysis |

## Governance

OLOS is stewarded by the **OLOS Architecture Working Group**. The specification proposes a Rotating Consortium Custody model for the Trust Registry at launch — founding participants jointly operate registry keys via threshold signatures — with a published path toward independent-foundation governance (in the style of EMVCo or the Linux Foundation) once a critical mass of independent merchants has onboarded. Registry operators are structurally barred from also operating settlement, so no single entity controls both trust attestation and asset netting.

## Where This Stands

This is an open specification at the **public proposal / technical review** stage — not a running network, sandbox, or certified implementation. The near-term goal is independent technical review from payments, security, and distributed-systems specialists, not a product launch timeline. A roadmap will be published once that review process is further along, rather than committing to dates now.

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